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Machine-learning-based prediction and optimization of emerging contaminants' adsorption capacity on biochar materials

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ABSTRACT

Biochar materials have recently received considerable recognition as eco-friendly and cost-effective adsorbents capable of effectively removing hazardous emerging contaminants (e.g., pharmaceuticals, herbicides, and fungicides) to aquatic organisms and human health accumulated in aquatic ecosystems. In this study, ten tree-based machine learning (ML) models, including bagging, CatBoost, ExtraTrees, HistGradientBoosting, XGBoost, GradientBoosting, DecisionTree, Random Forest, Light gradient Boosting, and KNearest Neighbors, have been built to accurately predict the adsorption capacity of biochar materials toward ECs in aqueous solutions. A very large data set with 3,757 data points was generated using 24 input variables (i.e., pyrolysis conditions for biochar production (3 features), biochar characteristics (3 features), biochar compositions (6 features), and adsorption experimental conditions (12 features)) obtained from the batch adsorption experiments to remove 12 kinds of ECs using 18 different biochar materials. The rigorous evaluation and comparison of the ML model performances shows that CatBoost model had the highest test coefficient of determination (0.9433) and lowest mean absolute error (4.95 mg/g), outperformed clearly all other models. The feature importance analyzed by the shapley additive explanations (SHAP) indicated that the adsorption experimental conditions provided the highest impact on the model prediction for adsorption capacity (41 %) followed by the adsorbent composition (35 %), adsorbent characterization (20 %), and synthesis conditions (3)%). The optimized experimental conditions predicted by the modeling were a N/C ratio of 0.017, BET surface area of 1040 m^2/g, content of C(%) contents of 82.1 %, pore volume of 0.46 cm^3/g, initial ECs concentration of 100 mg/L, type of pollutant (CAR), adsorption type (Single) and adsorption contact time (720 min).

1. Introduction

Emerging contaminants (ECs) such as personal care products, pharmaceuticals, endocrine-disrupting compounds, fungicides, and pesticides have been commonly used, which raised the presence of their unintended discharge into the environment [1,2]. The ECs have been detected in numerous environmental matrices including surface water, ground water, municipal wastewater and agricultural land [3]. The ECs can have hazardous effects on aquatic life, such as metabolic and sex alterations, disruption in biodegradation activity, and induction of antibiotic resistance genes in natural environment [4]. Hence, it is vital to eliminate these ECs from water resources. In comparison with the advanced oxidation processes [5], membrane separation [6], and bioremediation [7], adsorption is regarded as a robust, cost-effective, non-toxic, and efficient technology for the elimination of a wide range of organic/inorganic contaminates [8,9]. The utilization of green adsorbents such as biochar materials is favorable to decline the global carbon footprint, greenhouse gas emission and simultaneously effective

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towards the remediation of aqueous contaminates due to their excess surface functional groups, higher surface area, well-developed pore structure, and tunable porous structures [10–12]. Biochar is a carbon-rich material synthesized from the thermochemical treatment of biomass under a limited oxygen environment [12]. Since the old age production by slash-and-char method, biomass pyrolysis is extensively used to produce bioenergy and biochar. Pyrolysis conditions such as temperature, residence time, and heating rate have been reported to significantly affect the properties of synthesized biochar [13,14]. Thus, it is vital to select the optimized pyrolysis conditions to obtain highly effective biochar. Apart from the pyrolysis conditions, chemical activation such as alkaline, acid treatment, oxidation, or mineral impregnation is commonly applied to enhance the porosity, surface functionality, and surface area of biochar, leading to a higher adsorption performance [15].

For the adsorption of ECs on biochar materials, the mass transfer primarily involves chemical reactions (ionic bonding, covalent bonding) and physical interactions (hydrogen bonding, Van der Waals forces) among the ECs and biochars [16]. Hence, the adsorption capacity can primarily be affected by various factors including biochar synthesis conditions (e.g., pyrolysis temperature, pyrolysis time), biochar characteristics (e.g., BET surface area, pore size, pore volume), biochar compositions (e.g., contents of oxygen, carbon, nitrogen, ash, etc.), functionality and molecular structures of ECs molecules and experimental adsorption conditions (e.g., contact time, adsorbent dosage, initial ECs concentrations, solution pH, solution volume) [17,18]. The ECs adsorption using biochar had been extensively investigated in the literature using traditional adsorption statistical models [19–21], which indicate the potential adsorption mechanisms including electrostatic interaction, hydrogen bonding, pore-filling adsorption/partitioning, and π-π interactions. Nonetheless, these models are normally static models in the sense that they only explain the adsorption mechanism under a certain set of fixed operating conditions. Besides, the batch experimental technique is costly and time-consuming as they mainly rely on one parameter at a time, which might also fail to describe the inter-parameter interaction on the adsorption capacity. Hence, it would be desirable to build general explainable data-driven mathematical predictive models for studying ECs adsorption, gaining novel insights into the contribution of every single influencing factor and their collective impact on the ECs adsorption capacity.

Machine learning (ML) as a data-driven mathematical approach provides an outline to map complex and non-linear relationships among various inputs and output parameters of a process [22,23]. Especially, due to high-speed computation, self-learning capability, and simple design of algorithms, these predictive models provide optimum/precise solutions and could be considered instead of statistical models [24,25]. Until now, numerous simple ML algorithms such as gaussian processes, K nearest neighbors (KNN), and decision trees (DT) were successfully utilized in wastewater applications [26,27]. In KNN, the output is predicted using local interpolation of KNN from the query point [27]. In the DT algorithm, the target variable values are predicted by learning simple binary decision rules, which are incident from data features [28]. To boost the learning capabilities of DTs various ensemble techniques have been proposed such as random forest (RF), ExtraTrees (ET), and Bagging [29,30]. These ensemble techniques train numerous decision trees on random subsets of data training and then sum their predictions. This randomization declines alterations, enhancing prediction robustness and accuracy. Gradient-based training of decision trees has been planned to hasten their training process and improved prediction performance [31]. In this model, the gradient information of the loss function is applied to optimize the decision parameters. Numerous alterations to this technique led to the development of XGBoost, HistGradentBoosting (HGB), CatBoost (CB), and Light Gradient Boosting Machine (LGBM) algorithms [29,32]. These decision trees have various advantages such as being easier to understand, visualize, and interpreter. Nonetheless, the prediction of these decision trees can be biased, unstable, and can be overfitted on a small dataset. Besides, decision trees also need feature engineering to attain optimal performance [33].

Few literature reports applied ML modeling for wastewater data to predict the harmful impact on the environment, elimination performance, and quality of the water resources [34,35]. These publications have applied fewer ML models to predict the adsorption performance without clear evidence that the applied model is the appropriate choice for the given task [4,36]. Besides, they collect a small dataset (32–1750 data points) along with limited input variables on wastewater treatment from a limited number of publications [36–39]. Additionally, various ML models can also predict different performances, which led to a certain degree of alteration in the model prediction accuracy. Hence, the choice of an appropriate ML model for ECs adsorption from an aqueous solution is highly vital. To the best of our knowledge, this is the first study comparing ten different tree-based ML models for the ECs adsorption on biochar surface using a very large dataset (3757 data points) and 24 input features comprising various adsorption experimental conditions (12 features), biochar synthesis (3 features), biochar characteristics (3 features) and biochar compositions (6 features). Another key objective was to investigate the feature importance and impact of each adsorption condition and biochar property for ECs adsorption and classify the synergies among these input features on adsorption capacity. Besides, the accuracy of ML models was also addressed through the optimization of hyperparameters, which have normally been overlooked in ML studies for wastewater treatment. Finally, an engineering guideline for future screening and design of biochar materials for the adsorption of ECs under various conditions was also provided.

2. Experimental section

2.1. Methodology

Fig. 1 summarizes the schematic flow of the methodology to predict biochar adsorption capacity towards ECs and investigate the impacts of experimental adsorption conditions on capacity. As shown in Fig. 1, the first step of the current study is data collection. This process selected 24 parameters that includes biochar synthesis, biochar characteristics, biochar composition and adsorption experimental conditions consisting of 3,757 data points. In the second step, the ML models are trained and tested to specify the best models for predicting adsorbent capacity. During the process, the 10 ML models structured with optimized hyperparameters are compared. Afterward, prediction and feature analysis based on models are performed. The model evaluated with the highest predictive performance estimates the capacity of biochar from the wastewater and adsorption conditions. In addition, the post-processing techniques, such as shapley additive explanations (SHAP) and partial dependence (PD), interpret the causes of decisions made by models in detail.

2.2. Model development and evaluation

2.2.1. Data preparation

Herein, the dataset, consisting of 3,757 data points, was created by stacking our own experimental results for biochar synthesis and their adsorption performance towards the adsorption of ECs (e.g., ibuprofen, carbamazepine, alachlor, etc.). All these experiments were conducted between 2019 and 2022. The summary of the adsorption experimental conditions, including target ECs, materials feedstock, and process optimization conditions, is provided in Table S1. Besides, detailed information about biochar and pollutants: biochar production, preparation process for pollutants, biochar characteristics, detailed adsorption conditions, production equipment, and analytical methods were presented in our previous reports [10,40–45]. The input variables contain 24 parameters that significantly influence ECs adsorption, and can be divided into biochar synthesis conditions, biochar characteristics, biochar

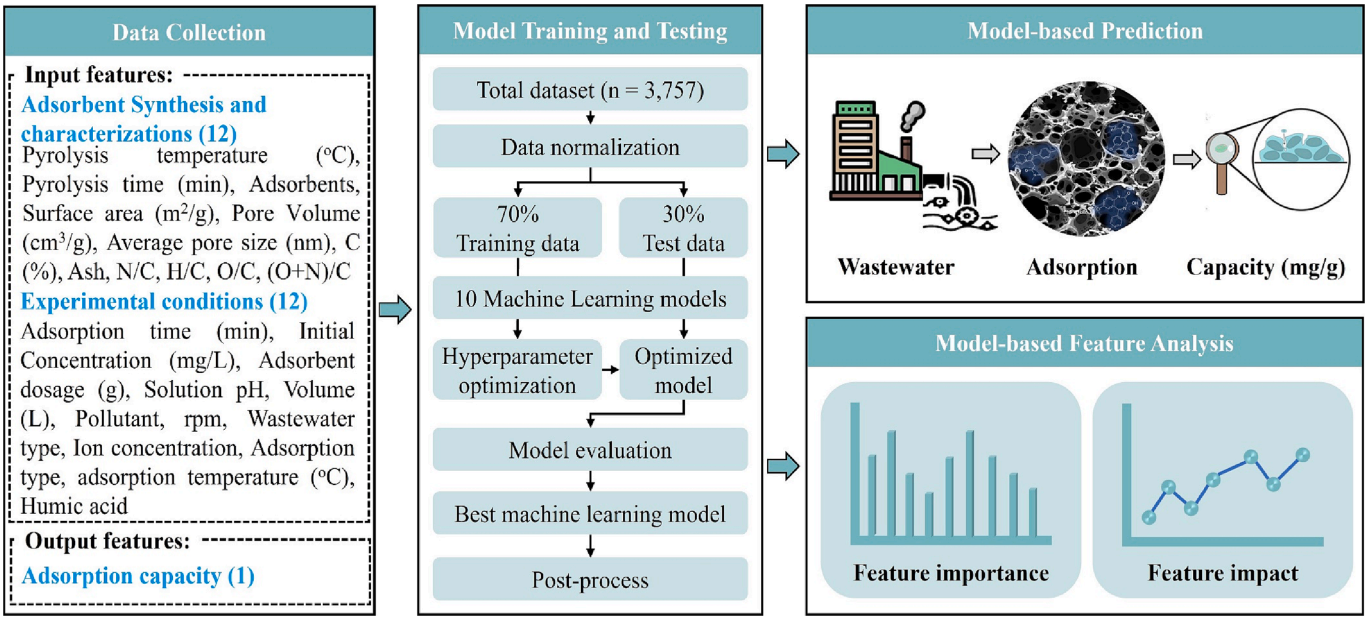

Fig. 1. Schematic illustration of the entire study.

compositions, and adsorption experimental conditions. The biochar synthesis conditions include pyrolysis temperature (℃), pyrolysis time (min), and synthesized biochar. The biochar characteristics consisted of surface area (m^2/g), pore volume (cm^3/g), and average pore size (nm). The biochar compositions measured C (%), Ash (%), and ratios of N/C, H/C, O/C, (O + N)/C. The experimental adsorption conditions included adsorption time (min), adsorption temperature (℃), initial ECs concentration (mg/L), adsorbent dosage (g), solution pH, solution volume (L), pollutant, revolutions per minute (rpm), wastewater type, ion concentration (M), and adsorption type. The output variable was adsorption capacity (mg/g). Eq. (1) was used to calculate the adsorption capacity,

$$Capacity = \frac{(C_i - C_f)}{m} \bullet V \tag{1}$$

Where C_i and C_f indicate the initial and final ECs concentration (mg/L), respectively. V and m are the volume of solution (ml) and the amount of adsorbent used (g).

In addition, the collected data went through three data preprocessing before being applied to the ML models. As our experiment-based datasets had no missing or erroneous value, standardization of numerical data was performed from the beginning without any steps for data quality improvement. Standardization of the dataset is a crucial requirement for ML models to eliminate the possibility of errors due to different scales of input features during training [46]. Subsequently, four categorical input variables, such as synthesized biochar, pollutant, wastewater type, and adsorption type, were assigned numerical attributes with a one-hot encoder. The encoder converted each category value into separate columns and gave either zero or one value to the columns. This process allows categorical data to be applied to machine learning algorithms as unordered numerical values. Additionally, it is essential because ML algorithms cannot use input values other than real numbers [47]. Finally, the preprocessed input data is randomly split at a ratio of 7:3. The 70 % of the data and the remaining 30 % were used for training and testing of the model, respectively. The data composition used in the training and testing process is designed to be the same for all ML algorithms for performance comparison under the same conditions.

2.2.2. *Categorical boosting model*

This study evaluated the adsorption capacity prediction model with the best performance by applying 10 different tree-based ML models. The applied ML models includes bagging (B), CatBoost (CB), DecisionTree (DT), ExtraTrees (ET), HistGradientBoosting (HGB), XGBoost (XGB), GradientBoosting (GB), Random Forest (RF), Light gradient Boosting (LGBM), and KNearest Neighbors (KNN). The tree-based models train patterns in given data by successively dividing data into subgroups using decision rules [48]. The criteria and combination of decision rules significantly impact the performance of models. Therefore, the models applied in this study are algorithms derived from research results investigating advanced tree combination methods. Among the models, the highest-performing model, the CB model, is described here. Information on remaining nine models is provided in the supplementary information (SI) file. Moreover, the supplementary material also includes optimized hyperparameters for all ten models.

The CB model, categorical boosting, was initially developed by Yandex, Russia, as one of the gradient-boosting decision trees (GBDT) and has been applied in various fields, such as recommender systems and particulate matter forecasts [49,50]. Recently, the high achievement of the model was reported for predicting chemical reactions, for example, photocatalytic performance and compressibility of ionic liquids [24,42]. The GBDT algorithms attempt to construct a high-performance model by sequentially fitting a new DT that complements the previous DT using the residuals and then linearly combining them. The ensemble method adopted by GBDT dramatically improves the data pattern learning ability. However, the slow learning speed due to a large amount of computation and the overfitting that occurs in reducing the residual are pointed out as disadvantages. Therefore, the CatBoost adopts a discriminate learning strategy to overcome the limitations of existing GBDT. The time-consuming problem is improved by forming a symmetric tree structure while creating every new tree. The random permutation, which randomly divides the entire given data into folds and applies boosting techniques to the data belonging to each fold, aid in preventing overfitting problems. Furthermore, CB models implemented ordered boosting, which computes residuals by assigning an order to each fold during the data training process. The ordered boosting approach can improve the issue of training with duplicate data, thus avoiding the potential for excessive time consumption and overfitting. Also, the CB model contains a specialized algorithm for categorical feature processing. The algorithm groups categorical data into a limited number of clusters and then applies one-hot encoding subsequently. As a result of those procedures, the CB models work explicitly for data with high cardinality.

2.2.3. Performance metrics

The predictive performance of the developed model was quantified by statistical parameters, including mean absolute error (MAE) and coefficient of determination (R^2). The MAE is an intuitive indicator of model error. A perfect model has a MAE value of 0; the closer the value is to 0, the more accurate the model is. The R^2 provides the degree to which the estimated value of the model fits the actual observed value. For the R^2, a value close to 1 indicates high performance. The mathematical equations of each metric are given below:

$$MAE = \frac{\sum_{i=1}^{N}(O_i - P_i)}{N} \quad (2)$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(O_i - P_i)^2}{\sum_{i=1}^{N}(O_i - \overline{O})^2} \quad (3)$$

Where N is the number of observations. O_i and P_i indicate the observed and predicted value, respectively. $\overline{O}$ indicates the mean of observed data. To further investigate that the 10 included models were not dependent on data splitting. Herein, the entire 3757 data points were randomly split into training and test dataset using 7:3 for 1000 times. For each time the models were trained and tested. The distribution of calculated test R^2 and MAE values were plotted in the form of a violin plot.

3. Result and discussion

3.1. Descriptive statistics

The data density and distribution of the biochar synthesis, physical properties, and composition of biochar were compiled in the form of box plot, as presented in Fig. 2. Box plot shows the descriptive analytics in terms of minimum, maximum and median values to gain a deep insight into the raw data. For biochar synthesis, the pyrolysis temperature and time were mainly concentrated around 800 °C and 120 min, respectively. The BET surface area of biochar is a vital factor affecting the adsorption capacity. In the current study, the surface area of biochar is mainly concentrated in the vicinity of 427.5 m^2/g, and a few biochar materials can obtain surface area higher than 1950 m^2/g. The pore volume and average pore size of biochar were intense around 0.293 cm^3/g and 4.9 nm, respectively. Apart from the physical characteristics, the presence of surface functional groups such as amino, hydroxyl, and carboxyl on biochar surfaces can significantly influence the adsorption performance [38]. The relative elemental contents (C, H, O and N) in biochar were varied because of the various types of feedstocks and pyrolysis conditions. The contents of carbon varied from 68 % to 88 %, representing the degree of carbonization. The polarity index (O + N/C) was applied as an indicator of the polar surface functional groups concentrations [51], which was concentrated around 0.053. The O/C molar ratio can be applied to measure the abundance of oxygen-based surface functional and hydrophilicity of biochar, while the aromaticity of biochar was represented by the molar ratio of H/C [37]. The median values of O/C and H/C molar ratios in the current study were around 0.03 and 0.04, respectively. The nitrate surface contents were denoted by the N/C molar ratio, which was mainly concentrated around 0.03. The ash contents in biochar can substantially alter the physiochemical properties of biochar and the organic matter distribution [52]. The role of ash in the adsorption of pollutants was controversial [53,54]. The ash contents ranged between 5.5 % and 24 % in the current data. The variation in the physicochemical properties of biochar was because of the difference in the raw materials, pyrolysis temperature and pyrolysis time. All these

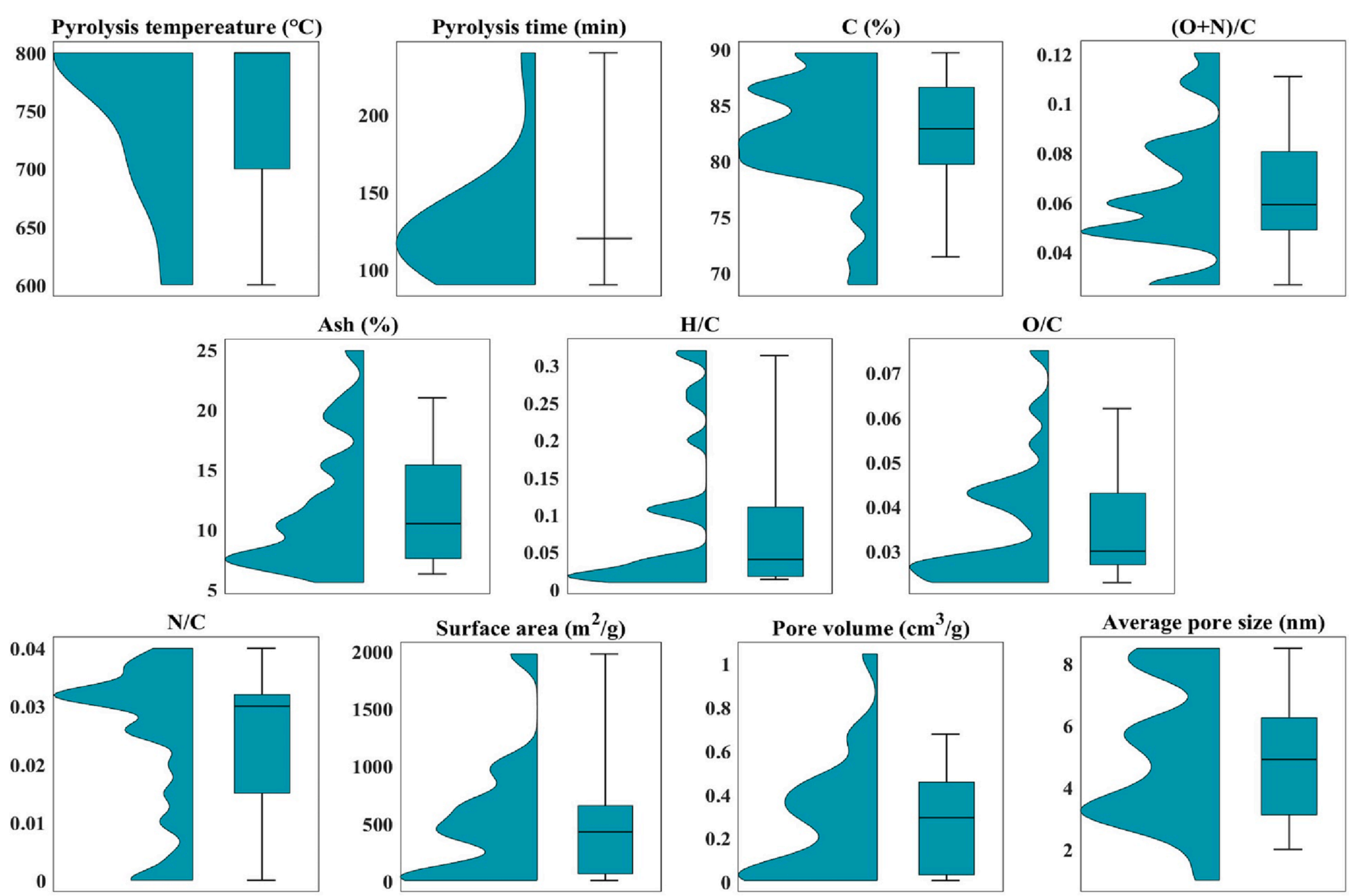

Fig. 2. Boxplot of input parameters related to biochar synthesis, characterizations and compositions.

factors were overlapping; thus, we have built the ML models by mixing all the biochar materials based on their physicochemical characteristics and adsorption performance.

Fig. S1 displays the Pearson correlation (PC) matrix among various input features. A strong positive correlation was witnessed in the case of biochar compositional characteristics (such as (O + N/C), H/C, O/C, and N/C). The correlation values for biochar composition variables were higher than 0.6, suggesting a stronger interdependence. Besides, surface area and initial pollutant concentrations were the only input features in physical properties and adsorption reaction conditions that have a considerable effect on adsorption capacity. Apart from the aforementioned input features, no significant correlation (PC values ranged between − 0.3 and 0.3) was recorded among the remaining inputs. The weaker correlation among most of the input features suggests retaining all of them during the predictive model building, as every feature would individually contribute to the model performance.

3.2. Model performance evaluation

Ten tree-based ML models, named CB, ET, HGB, XGB, GB, DT, RF, B, LGB and KNN were developed and evaluated to predict the adsorption capacity of ECs on biochar surface using 24 input features explained in the data preparation section. All of these models were trained using 2630 data points and the hyperparameters were optimized via Bayesian optimizer. A detailed discussion on hyperparameter optimization was also included in the supplementary information (Section S2.0). Once the selected ML models were trained and their hyperparameters were optimized, then, the prediction accuracy was evaluated using the regression plot. Fig. 3 displays the regression plots and ridge plots for experimentally calculated adsorption capacity plotted against the predicted values determined for the training and test datasets by various ML models. The ridge plots situated around the X and Y axis represent the model data used for training (70 %) and test (30 %) datasets. The blue solid lines represent the training, while the red dotted lines are for the test samples. The black dotted lines showed the perfect regression line (i.e., y = x). The regression plots showed that the prediction accuracy of all ten selected models was comparable, yet the CB model exhibited the most accurate prediction for the test data (Fig. 3(a)). Fig. 4 illustrates the residual plots and their normality distribution of training and test dataset using the ten selected models. The residual is defined as the difference among the observed (Y) and predicted ($\hat{Y}$) valued [55]. A model is considered more reliable, if the residual errors were situated on or close to the zero line (black line in the given figures) and possess normal distribution. The results showed that most of the residual errors and their histograms were situated near the zero line in the case of CB, ET, HGB, and XGB models. On the other hand, LGB and KNN exhibited more scattered residual errors. Besides, the residual errors were still situated close to the zero line upon an increase in the predicted values, which suggests the outstanding performance of the developed models. Fig. 5.

The accuracy of these models was further compared in terms of matric such as R^2 and MAE, as presented in Fig. 4(a and b), respectively. The test R^2 values for CB, ET, HGB, XGB, GB, DT, RF, B, LGB and KNN models were 0.9433, 0.9426, 0.9424, 0.9419, 0.9417, 0.9415, 0.9386, 0.9384, 0.938, and 0.8979, respectively. It was clearly seen that the CB model had relatively better prediction performance than the remaining nine models, which further confirmed by the fact that the CB model also had the lowest test MAE values of 4.95 mg/g among all the ten models. Overfitting was not an issue, as both training and testing R^2 values were approximated. It implies that the dataset can accurately predict the adsorption capacity upon comparison with other datasets. Based on the residual plots, performance metrics and regression plots results, the CB model was chosen as the most precise and reliable model, thus used for further post-processing investigations.

Table 1 compares the R^2 value of the current study with the literature reports on adsorption performance prediction towards various aqueous pollutants. As evident, the number of data points and input features used in the current study is remarkably higher than in the other reports suggesting the diversity of the considered experimental conditions [4,36,37,39,56]. The slightly less R^2 in the current study than the few other reports could be due to the nature of the data applied, i.e., diversity of adsorbents, number, and type of input features, number of data points used to build these models, and type of build algorithms. Nonetheless, the R^2 value was higher when compared with the studies that used more than 1000 data points. The ML codes and raw data used to build these codes are also provided in the current study, which sufficiently improves the reproducibility and transparency of model results. Therefore, the prediction made in the current study is more realistic and suitable for actual wastewater treatment problems.

3.3. Feature importance analysis

Adsorption of ECs onto the biochar surface can be influenced by several factors, yet obviously, their impacts are different. Fig. 6 displays

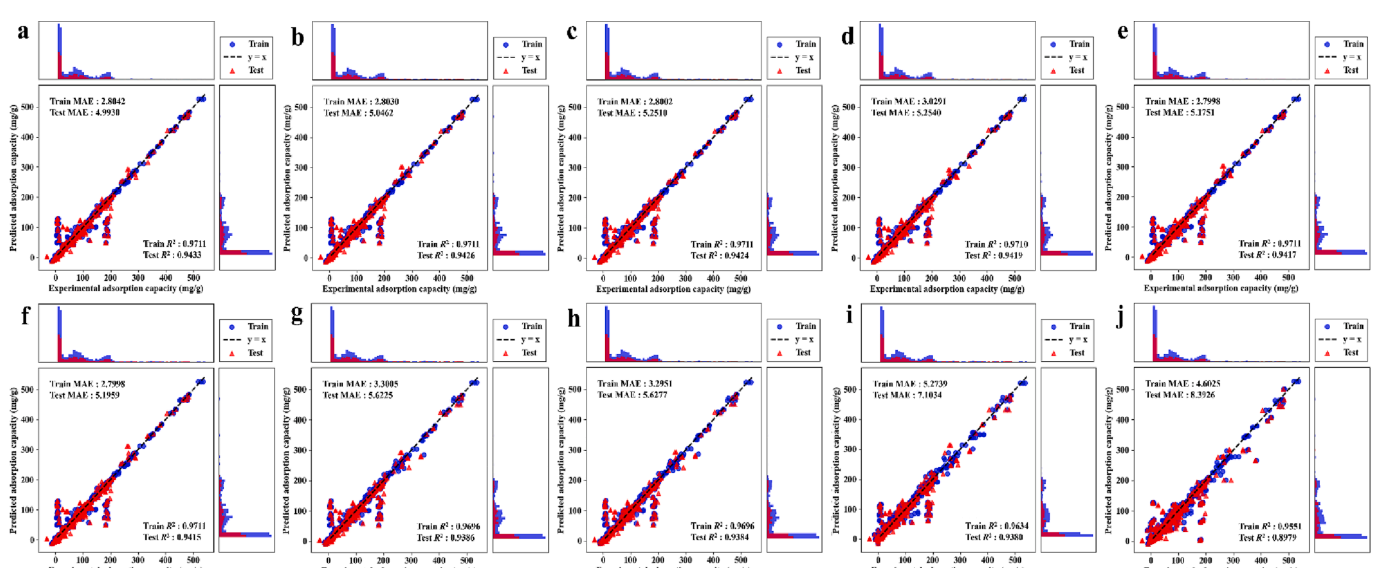

Fig. 3. Regression Plots and data distributions of experimental vs predicted adsorption capacity for (a) CatBoost, (b) ExtraTree, (c) HistGradientBoosting, (d) eXtremeGradientBoosting, (e) GradientBoosting, (f) DecisionTree, (g) RandomForest, (h) Bagging, (i) LightGradientBoosting and (j) KNearestNeighbors models.

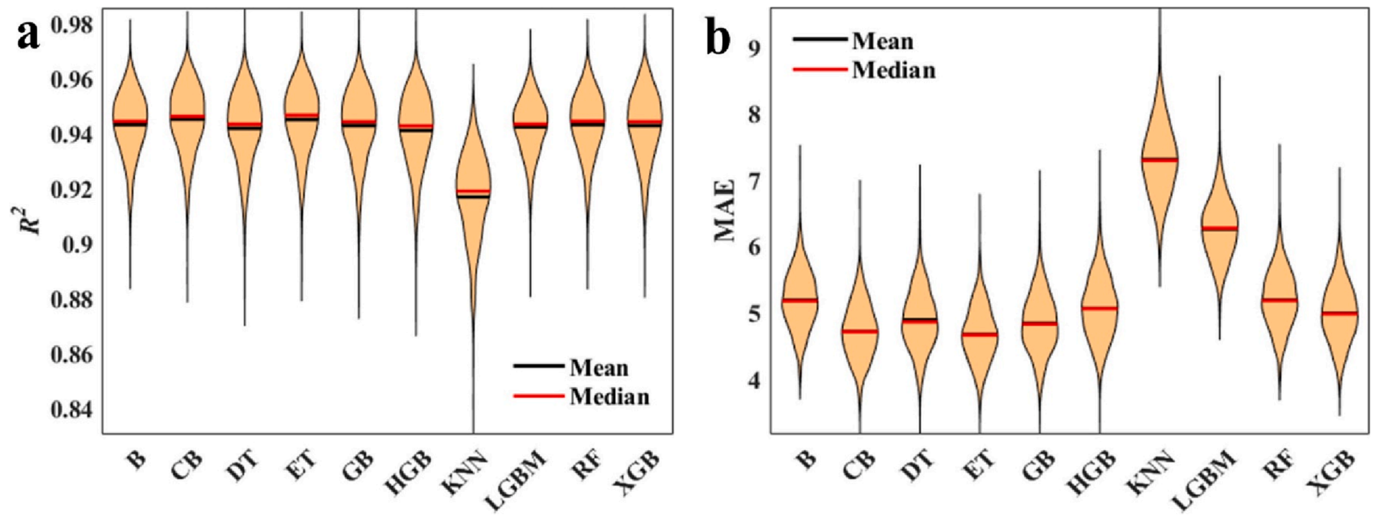

Fig. 4. Statistical assessment of ten selected model (a) R^2 and (b) MAE values.

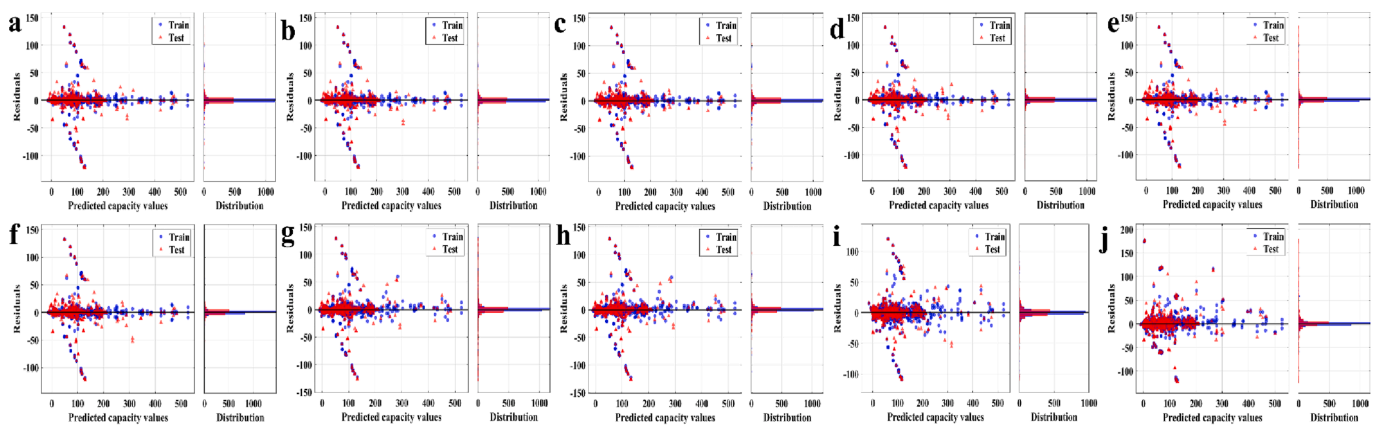

Fig. 5. Residual plots of experimental vs predicted adsorption capacity for (a) CatBoost, (b) ExtraTree, (c) HistGradientBoosting, (d) eXtremeGradientBoosting, (e) GradientBoosting, (f) DecisionTree, (g) RandomForest, (h) Bagging, (i) LightGradientBoosting and (j) KNearestNeighbors models.

Table 1
Comparison of the predictive strengths of various ML models for the adsorption capacity of various pollutants.

Pollutant	ML algorithms	Best model	Number of data points	Number of input features	R^2	ML Codes availability	Raw data availability	Ref.
Antibiotics	RF, GBT, ANN	RF	326	09	0.908	×	×	[4]
Neutral red	Multi-layer Perceptron, Passive aggressive regression, and Decision Tree	DT	32	2	0.99	×	×	[36]
ECs	Kriging model, ANN-LFER	Kriging model	1750	7	0.94	×	√	[39]
Heavy metal ions (HMI)	ANN, RF	RF	353	14	0.973	×	√	[38]
HMI	Linear regression, support vector regression, RF, ANN	ANN	777	10	0.97	×	√	[37]
perfluorooctanoic acid	AdaBoost, RF, Gradient boosting machine (GBM)	GBM	1343	21	0.878	×	×	[79]
Pharmaceuticals	KNN, RF, Cubist	KNN	170	10	0.92	×	×	[56]
ECs	CatBoost, ExtraTree, HistGradientBoosting, eXtremeGradientBoosting, GradientBoosting, DecisionTree, RandomForest, Bagging, LightGradientBoosting and KNearestNeighbors	CatBoost	**3757**	**24**	**0.943**	√	√	Current study

the feature importance of various input features to the adsorption capacity of ECs, which were investigated using the bar chart SHAP technique. Here, the "importance" is defined as the mean absolute SHAP value of all the points in the dataset. The input features were categorized into four types to investigate the influence of each type; experimental conditions, biochar characterizations, biochar composition, and biochar synthesis conditions. Insert of Fig. 6(a) shows the experimental conditions with 41 % showed the highest impact on the model prediction for adsorption capacity followed by the adsorbent composition (35 %), adsorbent characterization (20 %), and synthesis conditions (3)%).

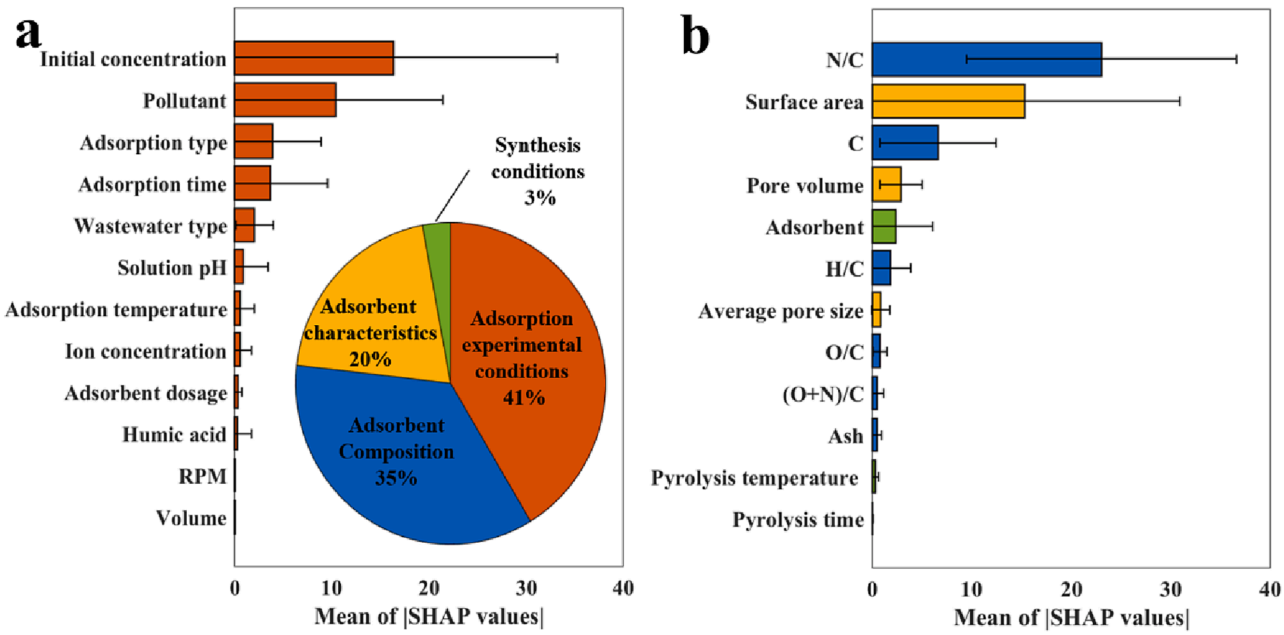

Fig. 6. Mean absolute SHAP values for feature importance analysis (a) adsorbent materials, (b) adsorbent experimental conditions, Insert of Fig. 6(b) displays the percent distribution of the feature importance among the adsorption experimental conditions, adsorbent characteristics, adsorbent composition and adsorbent synthesis conditions.

Among the experimental conditions, initial concentration had the highest influence, which reflects the quantity of the ECs in the reaction mixture (Fig. 6(a)). The effect of pollutants was also quite obvious, because various ECs pollutants possess different surface properties, thus affecting their interaction with biochar adsorbents. The adsorption type was the third most influential feature, which was related to the single and competitive adsorption properties towards ECs. In the adsorption experimental category, the concentration of Humic acid, agitation speed (rpm) and volume of reaction mixture were the least important input feature. Among the adsorbent composition, the ratio of N/C reflects the nitrate-based surface functional groups on biochar, and it was most influential in this category (Fig. 6(b)). Literature reports showed that an improvement in surface functional groups and the development of the *meso*/micro-porosities of biochar can substantially enhance the adsorption performance [57–59]. The surface area followed by the pore volume were the main contributors in the biochar characterization class, which was consistent with the common understanding of the adsorption mechanism [59,60]. The biochar synthesis conditions had a minute statistical significance on the model prediction with type of biochar on top of this category.

Fig. S2. Depicts the SHAP global feature importance of all 24 input features. The colors in the figure represent the values of the input features, where red shows high and blue means lower values. It means that the red on the left-hand side of the center line of the plot indicates a negative correlation with the predicted adsorption capacity, while red on the right-hand side of the plot suggests a positive correlation. Here, the N/C is clearly the most influential feature, followed by the initial ECs concentration, adsorbent's surface area, and various ECs, C, as expected. In the meantime, the initial concentration, surface area, adsorption type, adsorption reaction time, and pore volume are also positively correlated with the adsorption capacity. However, N/C, solution pH, and inorganic anion concentration were negatively correlated with the adsorption capacity. According to the SHAP analysis, the Ash contents, adsorbent dosage, and pyrolysis temperature were the least important feature among all 24. The feature importance analysis can also provide various clues to investigate the adsorption mechanism and can propose improvements in future biochar synthesis.

The additional investigation included analyzing a single data point with the highest adsorption capacity among the collected datasets using local SHAP explanations. The local SHAP plot, as shown in Fig. S3, quantified the influence of input variables on the prediction of the CB model. The x-axis of the graph represents the local SHAP values of individual input features, where each value denotes the contribution of the input feature to increase the adsorption capacity prediction from the mean value of 54.405 mg/g in our training data to 526.916 mg/g in this data point. Overall, Fig. S2 demonstrates that the predicted adsorption capacity is significantly higher than the average adsorption capacity of the training samples. The exact SHAP values, represented by the bar length, reveal that the initial concentration of ECs (49.702 mg/L), biochar surface area (1977.6 m^2/g), type of ECs (Carbendazim), adsorbent material (PS_{BOX-A}), C contents (82.1 %), N/C (0.016), and adsorption type (Single) are the more influential input variables on the model than the ratio of H/C (0.1), adsorption time (1440 min), and the collective sum of the remaining 15 input features. Furthermore, the ranking of variable importance changed compared to the rank assigned based on the mean value for the entire dataset. This suggests that the important variables vary depending on the range of adsorption capacity, and for high adsorption capacities, factors such as initial concentration or surface area are more significantly important than the composition of the adsorbent.

3.4. Critical role of the most influential input features

In addition to the SHAP feature importance analysis, the partial dependence plots (PDPs) were performed to understand the contribution of the nine most vital features to the prediction of the adsorption capacity of biochar towards ECs. Fig. 7 displays the one-dimensional (1D) PDPs signifying the impact of single feature on the model behavior. The experimental values were illustrated on the x-axis, while the light gray color column denotes the experimental data distributions. According to the SHAP analysis, the N/C ratio was the most influential input feature, which controls the ECs adsorption. The excess functional groups on biochar surface (e.g., $-NH_x$, $-OH$) could experience chemical interaction with the functional groups on the ECs surface [61]. Nonetheless, a decline in the PDP value was seen upon an increase in the N/C ratio value (Fig. 7(a)), suggesting a negative influence on the adsorption

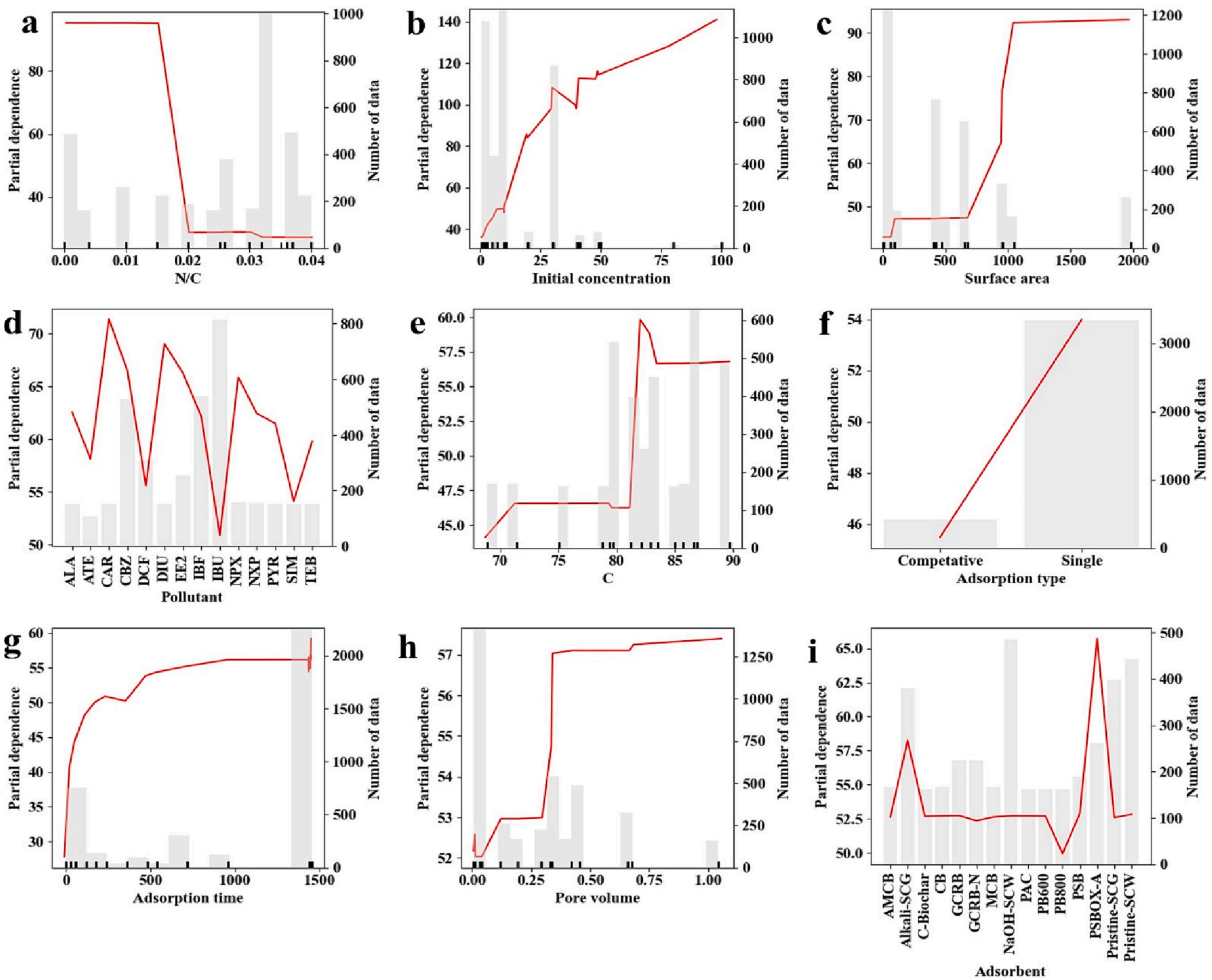

Fig. 7. SHAP dependence plot for nine most influential features with the highest SHAP values (a) N/C, (b) initial ECs concentration, (c) surface area, (d) pollutants, (e) C(%), (f) adsorption type, (g) adsorption time, (h) pore volume and (i) type of adsorbents.

capacity value. This can be credited to the competition between the formation and enrichment of *N*-containing functional groups in the biochar by partial dissolution in the aqueous phase by rigorous hydrolysis of nucleic acid, proteins and conversion of imines in feedstock to *N*-heterocyclic components by ring closure and dehydration [12,14]. Besides, the excess of *N*-based surface functional groups of biochar might also induce the production of a water film barrier that hindered the availability of interior active sites and caused a declining trend in adsorption capacity prediction [62]. The *N*-based functional groups on biochar were situated in the form of $-NH_x$ on adsorbent surface, which could be differentiated in future ML models upon the sufficient availability of relevant data.

After N/C ratio, the initial ECs concentration was the second most influential input feature for the adsorption process. Fig. 7(b) illustrates a rapid increase in PDP values when the initial ECs concentrations were enhanced from 0 mg/L to 50 mg/L. Afterwards, the increase in the PDP values became slower, suggesting an achievement of potential adsorption equilibrium at a 100 mg/L initial ECs concentration. Ozdemir et al. [63] interpreted this impact as overcoming all resistances to mass transfers among the adsorbent phases and adsorbate by driving forces of higher pollutants' initial concentrations. Thus, a maximum PDP value was recorded at 100 mg/L initial ECs concentration. BET surface area was the third most influential input feature, which could change the adsorption capacity prediction of biochar materials. Fig. 7(c) shows an increase in the PDP values upon increase in BET surface area. It was possibly due to the reason that surplus accessible adsorption sites can improve the interaction among the ECs molecules and biochar surface, which could lead to the higher adsorption capacity [64]. The most significant influence of BET surface area on adsorption of ECs on the surface of biochar materials was also consistent with the PC metrics (r = 0.65) (Fig. S1). However, the increase in the PDP values became extremely slow beyond a certain point, which suggest that the disproportionate hunt of a higher surface area should be avoided during the synthesis of biochar materials for the treatment of ECs containing wastewater [65].

The fourth most influential feature in predicting the adsorption capacity of ECs was the type of pollutants, as illustrated in Fig. 7(d). The results suggested that the higher partial dependence value was recorded upon applying the biochar towards the carbendazim (CAR) followed by diuron (DIU) and naproxen (NPX), while the lower partial dependence was recorded for ibuprofen (IBU), diclofenac (DCF) and simazine (SIM). The CAR, DIU, and NPX are known to have higher hydrophobicity and lower water solubility than the IBU, DCF, and SIM. Literature reports suggested that Hydrophobic compounds tend to have higher adsorption affinity towards biochar materials, which have a high surface area and hydrophobic surfaces [66,67]. Therefore, it is possible that the higher hydrophobicity of CAR, DIU, and NPX contributed to their higher adsorption performance on biochar materials. Additionally, the surface chemistry of biochar materials can also affect the adsorption performance of ECs. Biochar materials with a limited surface functional group, such as -NH_x, C=O, and –OH groups, have been shown to have higher adsorption capacity for non-polar pollutants, such as CAR, DIU, and NPX. A similar prediction was proposed in the PDP plot of N/C (Fig. 7(a)), where a higher number of surface functional groups led to a decline in PDP value. Thus, the higher adsorption capacity for CAR, DIU, and NPX compared with IBU, DCF, and SIM may be due to the combination of factors, including the hydrophobicity of the compounds and the surface chemistry of the biochar materials.

The presence of carbon contents in biochar is another important feature in the adsorption capacity prediction towards ECs. The C(%) contents exhibits the presence of C—C and C=C groups on the biochar surface. According to the PDP plot of C(%), the partial dependence was slightly enhanced and gradually achieved stability up to a certain amount (C% = 68–82 %) (Fig. 7(e)). A further increase in the C(%) contents on biochar surface from 82 to 84 % would lead to a rapid increase in partial dependence, and a slight decline was witnessed beyond that. The higher PDP values for excess carbon contents could be due to the reason that more carbon in biochar leads to a higher surface area and pore volume, which can result in a higher adsorption capacity [13,66,67]. These findings were also consistent with the adsorption study of other organic pollutants on biochar [68,69].

The type of adsorption can also highly influence the adsorption performance, as illustrated in Fig. 7(f). It can be seen that the single molecule adsorption exhibited a better partial dependence value than the competitive adsorption because it allows a more controlled and predictable process. In the competitive adsorption process, the outcome can be influenced by a number of factors such as the relative affinity of the molecules for the adsorption sites, the ratio of the molecules present, and the order in which they were adsorbed [66]. These variables can make it more difficult to predict the outcome of the adsorption process and may lead to less favorable results.

The contact time was the next important factor and its PDP impact on the adsorption capacity was also investigated, as illustrated in Fig. 7(g). As can be seen, the PDP value was first increased rapidly during the initial 480 min of the adsorption experiment. Then slow down and equilibrium was achieved approximately after 960 min. The rapid increase in the PDP value during the initial stages of the adsorption experiment could be credited to the higher BET surface area and excess number of surface-active sites that immediately capture the ECs molecules. Upon prolonging the adsorption time, the accessibility of ECs molecules to unoccupied active sites on the biochar surface declined and ultimately these active sites became saturated when the equilibrium was established. The equilibrium state is derived from diminishing the driving forces and saturation of the vacant surface-active sites [70]. The equilibrium state is also known as the situation when the ECs concentration in the solution achieves the dynamic balance with the interface. Depending upon these findings, 720 min could be selected as the suitable adsorption contact time.

Pore volume was the sixth most important feature predicted by the SHAP analysis. Fig. 7(h) depicts an increasing pore volume can highly enhance the partial dependence of adsorption capacity of biochar, yet when the pore volume was higher than 0.33 cm^3/g, the biochar has no obvious impact on adsorption performance. The increased partial dependency upon improvement in pore volume value can credited to the generation of new internal surface area within a biochar material, and the more pores a biochar material has, the more internal surface area it has available for ECs adsorption. Another reason for enhanced PDP value upon increasing pore volume might be the bigger pore size, which can easily facilities excess EC molecules during adsorption process [71]. However, if the pores are too large, the material may not be able to effectively adsorb smaller EC molecules, leading to a decline in adsorption capacity.

The type of adsorbent was the last and ninth most influential feature in this study. Fig. 7(i) shows the partial dependence of various biochars synthesized from various agricultural and food waste towards the adsorption of ECs. Among all the applied biochar, the $KMNO_4$-modified peanut shell biochar ($PSB_{OX\text{-}A}$) showed the maximum partial dependence, which suggests that it has the best adsorption capacity towards ECs removal. The higher adsorption capacity of $PSB_{OX\text{-}A}$ could be credited to the enriched surface functional groups, higher BET surface area, pore volume, and their interaction with the ECs molecules [72,73]. On the other hand, the lowest PDP value was recorded for NaOH pretreated pine sawdust biochar pyrolyzed at 800 °C (PB-800). The lower PDP value could be attributed to the formation of highly crystalline structures at higher pyrolysis temperatures, leading to the declined porosity and surface area [74]. NaOH treatment can be another reason for declined adsorption performance, as alkali-treated biochar can produce new surface functional such as carboxyl (COOH) and hydroxyl (OH) groups. These groups can compete with other functional groups on the biochar surface for adsorption sites, leading to a reduction in the overall number of active adsorption sites available [75]. Similarly, the other alkaline-treated biochars considered in the current study such as NaOH-SCW, PB600, and Alkali-SCG also showed relatively lower PDP values.

The relationship between the most important feature (N/C) and other influential numerical input features was investigated using a two-feature PDP plot, which is further helpful to visualize the interaction among them and optimize the predicted adsorption capacity (mg/g). The findings are illustrated in Fig. 8. The findings suggested that the dependence of adsorption capacity on initial concentration was stronger at lower N/C values (Fig. 8a) because the excess of the *N*-containing surface group might release to the aqueous solution and complete the ECs molecules for the adsorption on the active sites of biochar surface. The adsorption capacity was almost independent of the surface area of biochar when the N/C value was lower than 0.020 (Fig. 8b). The carbon contents lower than 82 % had the least impact on adsorption capacity, while higher carbon contents sufficiently improve the adsorption capacity at the lower N/C values (Fig. 8c). The interactions among the *N*-containing functional groups and adsorption time were also investigated (Fig. 8d), showing that an intermediate adsorption time is cost-effective and ideal to obtain a higher adsorption capacity at lower N/C value. The dependence of adsorption capacity on pore volume was more noticeable and potent again at lower N/C values (Fig. 8e), which was consistent with the other literature reports [4,37]. Finally, the synergistic effects among surface functional groups were also illustrated (Fig. 8f), suggesting that O-based surface functional groups (e.g., hydroxyl, carboxyl, etc.) were more favorable than that of the *N*-based surface group [76].

Overall, the PDP analysis was helpful for comprehending biomass conversion to biochar and allowing optimization in aqueous environmental purification applications. For instance, N/C was the most important input feature for biochar characterization and higher adsorption performance. In environmental purification applications, the PDP can be applied as a reference for improving the biochar quality for the adsorption treatment process, while balancing energy expenses to raise the temperature during pyrolysis. The current study offered a cost-effective practical route for further in-depth investigations of biomass conversion into biochar and their wastewater treatment applications.

3.5. *Implications and drawbacks of the current study*

CB model developed in the current study exhibited two key environmental applications. First, it offers deeper insightful information on the removal mechanism of ECs using biochar as adsorbents. The adsorption process contains several mechanisms, in which various influencing features could contribute to adsorption capacity including adsorption experimental conditions (initial concentration, contact time,

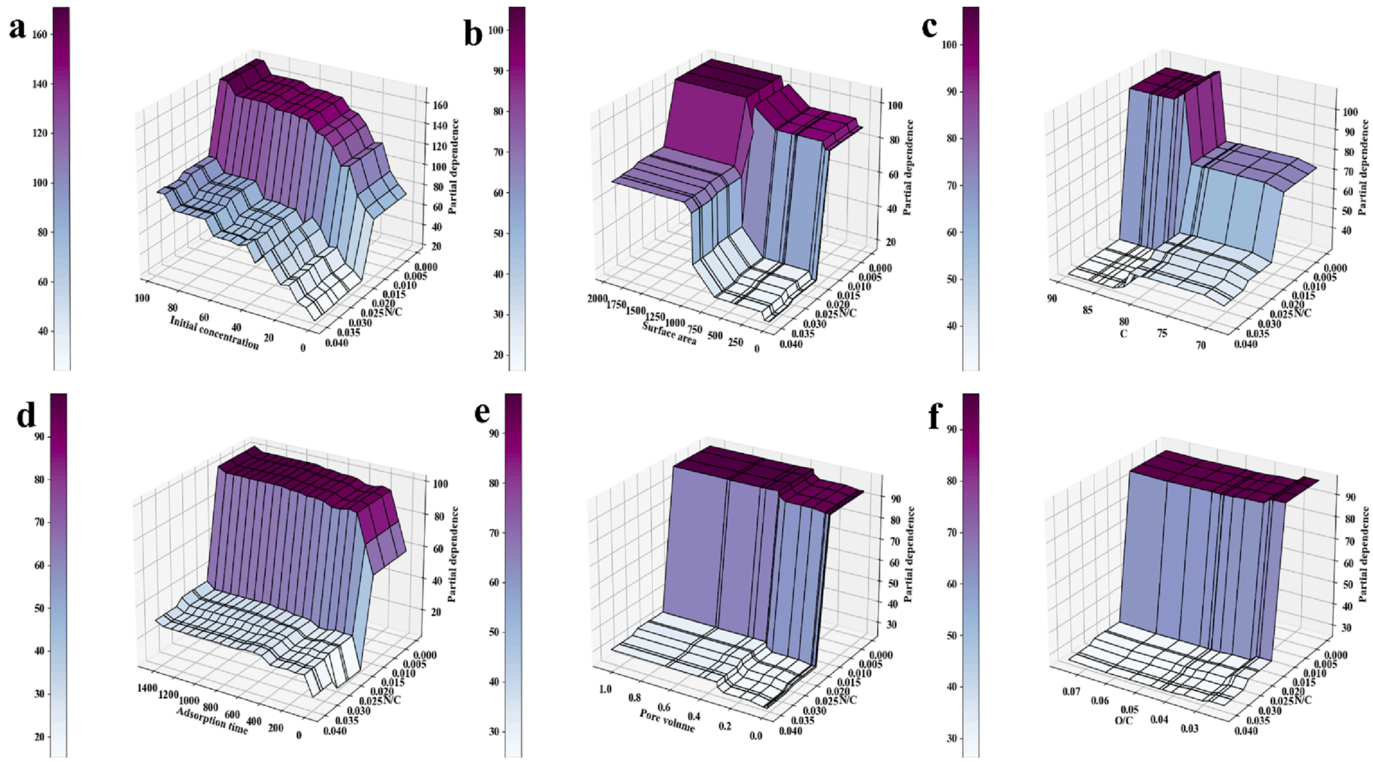

Fig. 8. Two-feature PDP dependence of ECs removal on biochar materials and the interaction among the two features. (The x-axis contains the top most influential numerical input features, while on the y-axis N/C is constant).

solution pH, etc.), biochar characteristics (BET surface area, pore volume, etc.,), biochar compositions (Ash contents, surface functional groups, etc.,), and biochar synthesis (pyrolysis temperature, pyrolysis time, etc.,). Few proposed assumption models or various experiments were unable to elucidate the adsorption mechanisms because of the higher uncertainty, which could be caused by the lack of advanced measurements or lack of abundant materials assessments. Second, these findings devise a potential technique to precisely predict the adsorption performance of biochar without performing a single experiment, which will be extremely helpful in actual wastewater treatment using biochar. The robustly build CB model can assist environmental engineers to predict or visualize the adsorption capacity of the given biochar without any time consumption, no new experiments, and higher generalizing capability. Besides, we already predicted the optimized adsorption process conditions, this will be highly helpful to accurately adjust the real experimental conditions. The findings also proposed that the adsorption capacity could be improved by optimizing the biochar composition, probably by adjusting the pyrolysis temperature and time.

Despite the outstanding performance of the developed tree-based ML models, there were still some limitations, which should be targeted in future studies. First, the adsorption capacity can be significantly affected by various other factors, such as raw materials, distinct surface functional groups, and distinct characteristics of target pollutants. Thus, it is suggested to consider it in future ML model development. Besides, the role of some highly toxic competitor organic compounds such as amino acids, humic substances, proteins, and saccharides on the active sites with ECs should also be considered to optimize the practical applications of biochar towards real wastewater. There is a close relationship between the dataset and ML algorithm, thus the trial-and-error approach in finding the optimal model and best configuration of input data is also needed to implement to the certain adsorption dataset before training and tuning deeply any ultimate model. In the current study, we have compared most of the tree-based ML models, yet it is also suggested that the neuron-based models should also be analyzed. Additionally, as the multicollinearity of independent variables can degrade prediction performance depending on the applied model or constructed data, subsequent studies should also consider the application of dimension reduction techniques such as feature selection and principal component analysis [77,78]. Next, prediction derived from data-driven ML algorithms may be difficult for water sector operators and environmental researchers to understand and apply. Thus, the friendly web user information or interface available is vastly desirable to develop along the predictive research.

4. Conclusion

Herein, ten tree-based ML models, CB, ET, HGB, XGB, GB, DT, RF, B, LGB and KNN were applied to predict the adsorption capacity of biochar materials towards the elimination of ECs. A largest of 3757 data point of ECs removal were generated under various conditions. The CatBoost model outperformed the remaining selected models based on the highest test R^2 and lowest MAE values. Residual plot showed that most of the errors were situated around the zero line. Additionally, the feature importance analysis suggested that the order of input features types were experimental conditions (41 %) > adsorbent characterizations (35 %) > adsorbent composition (20 %) > adsorbent synthesis conditions (3)%). The PDP was applied to accurately model the impact of nine most influential features on model prediction. Finally, a detailed discussion on model implications and drawbacks were also included, which could work as the basic guidelines for the wastewater treatment operators and environmental engineers.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The Python codes used to build the models and raw data used in this study are available at following GitHub link https://github.com/ZeeshanHJ/Adsorption-capacity-prediction-for-ECs

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2023.143073.

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